	FIR VEHICLE TESTING	19/01/2026
FIR No :FIR/Testing-Outdoor-UA/520700/2026/01229		
Part No/ Group :52070261R		
Subject :Report on fluid economy evaluation of Signa 3723.T BSVI-II Ashwamedh 5.6L (220 HP) engine, G950 (FGR:9.36) Transmission and 5.12 Rear axle ratio in comparison with Cowl 3723.T BSVI-II vehicle as a Base vehicle on JSR-Ranchi highway		
Test Vehicle	Project Name :Karnali Model Name : Signa 3723.T Karnali Chasis Number :PC53562032RJ00873 Engine Number :22L63919785	Enclosures : Annexure 1-5
FIR Status :	ESO	

Role	Name	Signature
Prepared By	Palash Das	19-Jan-2026
Checked By	Swapnil Jadhav	19-Jan-2026
Approved By	Ajaykumar Jadhav	19-Jan-2026

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA20 F222 (R4) March, 2021

Documents : Please click here

3723.T_FE_FIR_merged.pdf

Subject: -	Report on fluid economy evaluation of Signa 3723.T BSVI-II Ashwamedh 5.6L (220 HP) engine, G950 (FGR:9.36) Transmission and 5.12 Rear axle ratio in comparison with Cowl 3723.T BSVI-II vehicle as a Base vehicle on JSR-Ranchi highway

1.0 Conclusion:

1.1 Fluid economy in laden(rated load) condition at 40 kmph :

- Fluid economy of Signa 3723.T BSVI-II vehicle was found **0.807 Lit/ton/100km** which was inferior by **2.0 %** wrt Cowl 3523.T BSVI-II Base vehicle.

VI/ ENG — C

1.2 Fluid economy in laden(rated load) condition at 55 kmph :

- Fluid economy of Signa 3723.T BSVI-II vehicle found **0.863 Lit/ton/100km** which was inferior by **3.1 %** wrt Cowl 3523.T BSVI-II Base vehicle.

1.3 Fluid economy in Unladen condition at 40 kmph :

- Fluid economy of Signa 3723.T BSVI-II vehicle found **1.582 Lit/ton/100km** which was inferior by **4.4 %** wrt Cowl 3523.T BSVI-II Base vehicle.

VI/ ENG — C

1.4 Fluid economy in Unladen condition at 55 kmph :

- Fluid economy of Signa 3723.T BSVI-II vehicle found **1.759 Lit/ton/100km** which was which was inferior by **4.0 %** wrt Cowl 3523.T BSVI-II Base vehicle.

1.5 Overall fluid economy status at 40 kmph (80% Laden + 20% unladen condition):

- Fluid economy of Signa 3723.T BSVI-II vehicle found **0.895 Lit/ton/100km** which was inferior by **2.3 %** wrt Cowl 3523.T BSVI-II Base vehicle.
- Signa 3723.T BSVI-II doesn't meets the PALS target of "Leadership".

VI/ ENG — C

1.6 Overall fluid economy status at 55 kmph (80% Laden + 20% unladen condition):

- Fluid economy of Signa 3723.T BSVI-II vehicle found **0.961 Lit/ton/100km** which was inferior by **3.2 %** wrt Cowl 3523.T BSVI-II Base vehicle.
- Signa 3723.T BSVI-II doesn't meets the PALS target of "Leadership".

2.0 Result: Vide Observations

FAILURES CLASS:

A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.

B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.

C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.

D: Customer perception issues and minor irritants.

3.0 Test Vehicles: -

1. Signa 3723.T BS-VI OBD-II (ERCJ-1309)_Test vehicle
   Chassis no. - PC53562032RJ00873; Engine no. - 22L63919785.
2. Cowl 3523.T BSVI-II (ERCJ 1310)_Base vehicle
   Chassis no. - PC53532032RJ00877; Engine no. - 22L63919779.

Enclosures:

Annexure 1: Panel chart
Annexure 2: PAT sheet
Annexure 3: Duty cycle report
Annexure 4: Test request
Annexure 5: Test plan

Input Doc Ref.: - Test Request No :- Testing-Outdoor/UA/02766
DID No: – NA.
DDP No: – NA.

Work Order No. :- P.21.EA.456	Make :- TATA Motors Limited	T I No. – UA002748

Earlier Ref: - Nil.

4.0 Project Status:- DR3

5.0 Introduction:

As a part of RDE (OBD-II) Testing Program, TML Signa 3723.T vehicle fitted with M/s Cummins make Ashwamedh 5.6L BS-VI OBD-II (220 HP) engine, G950 (FGR:9.36) Transmission and 5.12 Rear axle ratio was offered for fluid economy evaluation along with TML Cowl 3523.T BSVI-II vehicle as a base vehicle. All vehicles were tested back-to-back at JSR- Ranchi-JSR route in laden and unladen conditions at 40 km/hr as well as 55 km/hr vehicle cruising speeds. The subject report will provide detail about fluid economy evaluation test carried out on all vehicles (Test vehicle, Base vehicle).

6.0 Testing:

Following test was conducted during evaluation :

Sr. no.	Test	Reference standard	Instruments used
1	Fluid economy evaluation	TST/VT/WG/158	Cummins Data Logger

7.0 Vehicle details:

Parameter		Base Vehicle	Test vehicle
Vehicle model		Cowl 3523.T (OBD-II) (ERC-1310)	Signa 3723.T (OBD-II) (ERC-1309)
Engine Model		Cummins Ashwamedh 5.6L BS-VI (220 HP) OBD-II	Cummins Ashwamedh 5.6L BS-VI (220 HP) OBD-II
Max Power		220 HP @2300 RPM	220 HP @2300 RPM
Max Torque		925 Nm @ 1000 ~1300 rpm	925 Nm @ 1000 ~1300 rpm
GVW (kg)		35000	37000
Gear Box		Model: Tata G950, Type: DD	Model: Tata G950, Type: DD
Gear Ratios.		1st gear:9.36, 2nd gear: 5.2, 3rd gear: 3.05, 4th gear: 1.84, 5th gear: 1.33, 6th gear: 1	1st gear:9.36, 2nd gear: 5.2, 3rd gear: 3.05, 4th gear: 1.84, 5th gear: 1.33, 6th gear: 1
Axle Ratio		5.12	5.12
Tyres	FA1	Size: 295/90R20, Make: M/s Bridgestone, Type :LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type :Non LCRR (Phase-I)
	FA2	Size: 295/90R20, Make: M/s Bridgestone, Type :LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-I)
	LA1	Size: 295/90R20, Make: M/s Bridgestone, Type :LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-I)
	RA1	Size: 295/90R20, Make: M/s Bridgestone, Type :LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-I)
	RA2	Size: 295/90R20, Make: M/s Bridgestone, Type :LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-I)
Tractive effort @ 1st gear		74851 N	81456 N
Tractive effort @ Top gear		7997 N	8703 N
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"	Type: E-Viscous fan and Size:24"

Parameter	Base Vehicle	Test vehicle
Vehicle model	Cowl 3523.T (OBD-I) (ERC-1160)	Signa 3723.T (OBD-II) (ERC-1309)
Base Cal Improvement (Cal Maturity %)	As per production released	95% Cal Maturity
Low Viscous Engine Oil (TMGO-10W30)	Yes (10W30)	Yes (10W30)
FE Switch	No (iUSFE)	No (iUSFE)
Engine Features (EOL : Enable/Disable)	Enable	Enable
ATS Optimization	Yes	Yes
APM	Yes	Yes
Cruise Control	No	No
Gear based VAM	No	No
Intelligent mode shifter	No	No
Moving Mass Estimation	No	No
Driveline Optimization	Yes	Yes
GSA (Gear Shift Advisor)	Yes	Yes
MSA (Mode Shift Advisor)	No	No
Low CRR Tyres	Yes	No
Low Viscous rear axle oil (80W90)	Yes	Yes
Smart Alternator (LIN)	NA	NA
THU/Unitised/Low friction bearing	No	No

8.0 Test result:

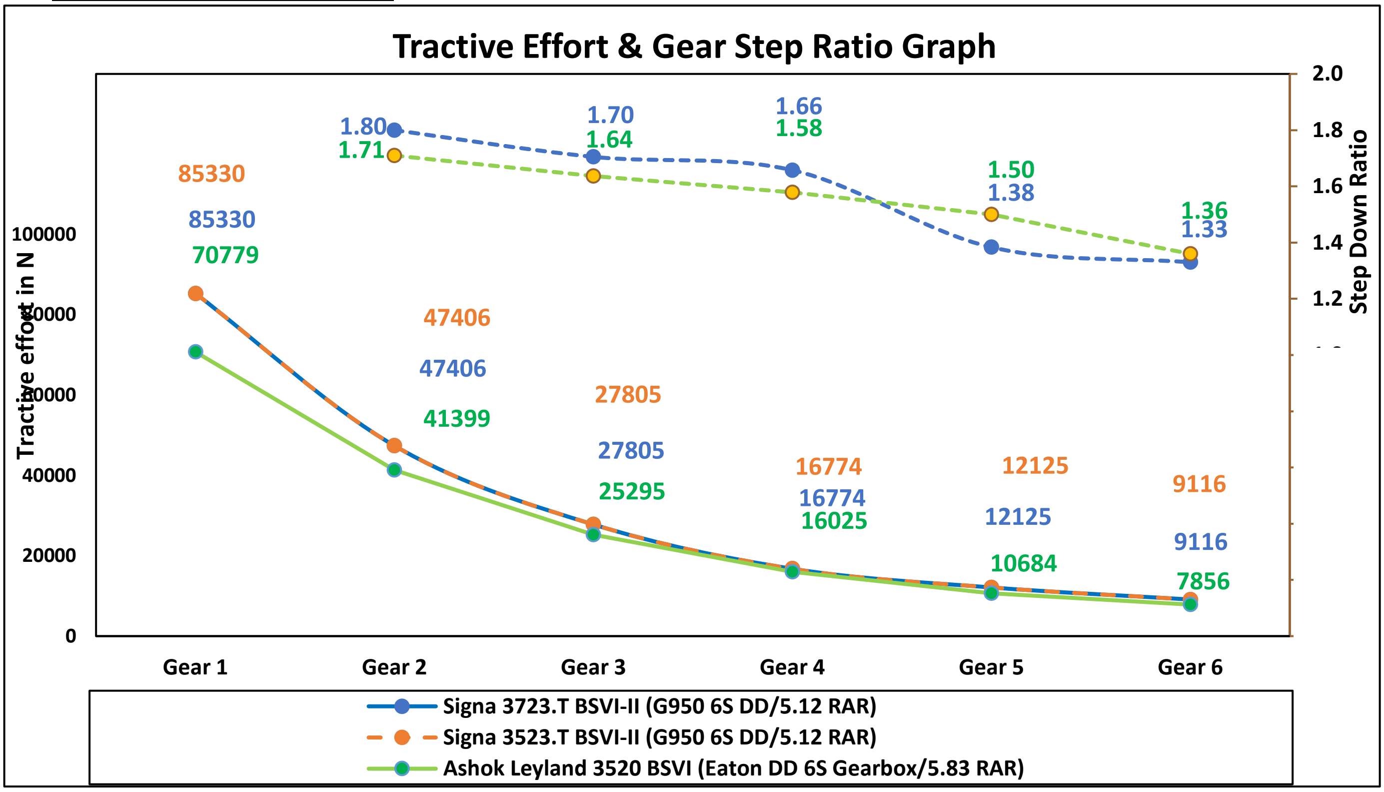
8.1 Cruising speed: 40 km/hr & 55 Km/hr

Sr. No.	Parameters			Base Vehicle : OBD-2	Test Vehicle : OBD-2
				Cowl 3523.T BSVI-II	SIGNA 3723.T BSVI-II
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1310	ERC-1309
2	Vehicle GVW/GCW		kg	35000	37000
	Unladen weight		kg	10000	10000
3	Engine Details (Make/cc/Norms/no. of Cyl.)			Cummins ISBe 5.6L/ Ashwamedh	Cummins ISBe 5.6L/ Ashwamedh
	Engine Max Power			220 HP @ 2300 rpm	220 hp @ 2300 rpm
	Engine Max Torque			925 Nm @ 1000-1300 rpm	925 Nm @ 1000-1300 rpm
	Engine Calibration Maturity Level			As per production released	95% Cal Maturity
4	Regeneration Strategy (Time ase/Soot base)			Time Based (100 Hrs)	Time Based (100 Hrs)
5	Gear Box (make/drive/FGR)			TATA G-950 DD	TATA G-950 DD
	Gear Ratios			1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00
6	RAR			5.12	5.12
7	Tyre Details			295/90 R 20 RIB Bridgestone Low-CRR	295/90R 20 RIB Bridgestone phase-I
8	LBSC	High & Low Engine Speed Breakpoint	RPM	2000 rpm, 1900 rpm	2000 rpm, 1900 rpm
		Vehicle Mass Threshold	kg	35000	37000
	VAM	Variable Acceleration 1, 2	kmph/sec	3.0,2.5	3.0,2.5
		Variable Acceleration Speed 1, 2	kmph	10, 40	10, 40
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	50,50	50,50
	RSG	Maximum Accelerator Vehicle Speed	kmph	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	120%	131%
		Torque Limit Maximum Adjustment Factor	Economy	130%	141%
	PTP	Powertrain Protection (Enable / Disable)		Disable	Disable
		Torque Limit	Nm	NA	NA
	USFE / Torque Table Revision (1/2/3/n)			USFE-1	USFE-1
9	Fan Details (type/make/blades)			E- 24'' High Efficiency Viscous fan	E- 24'' High Efficiency Viscous fan
FE Avenues (Engine Level)	Base Cal Improvement (Cal Maturity %)			As per production released	95% Cal Maturity
	Low Viscous Engine Oil (TMGO-10W30)			Yes (15W40CK4+)	Yes (15W40CK4+)
	FE Switch			No (iUSFE)	No (iUSFE)
	Engine Features (EOL : Enable/Disable)			Enable	Enable
	ATS Optimization			Yes	Yes
	APM			No	No
	Cruise Control			No	No
	Gear based VAM			No	No
	Intelligent mode shifter			No	No
	Moving Mass Estimation			No	No
FE Avenues (Vehicle Level)	Driveline Optimization			NA	Yes
	GSA (Gear Shift Advisor)			Yes	Yes
	MSA (Mode Shift Advisor)			No	No
	Low CRR Tyres			Yes	Yes
	Low Viscous rear axle oil (80W90)			Yes	Yes
	Smart Alternator (LIN)			NA	No
	THU/Unitised/Low friction bearing			No	No
FE Trial Details	Route			Jamshedpur-Ranchi-Jamshedpur	Jamshedpur-Ranchi-Jamshedpur

	Parameters	Unit	Base: Unladen Trial @ 40 Kmph		Base: Laden Trial @ 40 kmph		Base: Unladen Trial @ 55 Kmph		Base: Laden @ 55 Kmph		Test: Unladen @ 40 Kmph		Test: Laden @ 40 kmph		Test: Unladen Trial @ 55 Kmph		Test: Laden @ 55 Kmph	
FE Trial Details	FE Duty Cycle (Laden/Unladen)		Unladen Trial @ 40 Kmph		Laden Trial @ 40 kmph		Unladen Trial @ 55 Kmph		Laden @ 55 Kmph		Unladen @ 40 Kmph		Laden @ 40 kmph		Unladen Trial @ 55 Kmph		Laden @ 55 Kmph	
	Trial Date		14.12.2025	15.12.2025	07.12.2025	08.12.2025	12.12.2025	13.12.2025	09.12.2025	10.12.2025	14.12.2025	15.12.2025	07.12.2025	08.12.2025	12.12.2025	13.12.2025	09.12.2025	10.12.2025
	Trip Distance	km	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00	242.00
	Total Diesel Quantity	Litre	35.86	35.88	64.84	64.98	40.00	40.18	69.41	67.73	37.24	37.53	70.08	70.25	41.48	41.85	75.58	73.90
	Total Urea Quantity	Litre	1.60	1.60	4.10	4.30	1.70	1.70	4.30	5.10	1.80	1.80	4.00	4.40	1.60	2.00	5.60	4.70
	Equivalent Urea Quantity : 50% Consumption	Litre	0.80	0.80	2.05	2.15	0.85	0.85	2.15	2.55	0.90	0.90	2.00	2.20	0.80	1.00	2.80	2.35
	Total Eq. Fluid Consumed	Litre	36.66	36.68	66.89	67.13	40.85	41.03	71.56	70.28	38.14	38.43	72.08	72.45	42.28	42.85	78.38	76.25
	Urea uses % wrt. Diesel	%	4%	4%	6%	7%	4%	4%	6%	8%	5%	5%	6%	6%	4%	5%	7%	6%
	Fuel Economy	**kmpl**	**6.75**	**6.74**	**3.73**	**3.72**	**6.05**	**6.02**	**3.49**	**3.57**	**6.50**	**6.45**	**3.45**	**3.44**	**5.83**	**5.78**	**3.20**	**3.27**
	Status of Fuel economy	**%**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**-3.7%**	**-4.4%**	**-7.48%**	**-7.50%**	**-3.57%**	**-3.99%**	**-8.16%**	**-8.35%**
	Equivalent Fluid Economy	**kmpl**	**6.60**	**6.60**	**3.62**	**3.60**	**5.92**	**5.90**	**3.38**	**3.44**	**6.35**	**6.30**	**3.36**	**3.34**	**5.72**	**5.65**	**3.09**	**3.17**
	Status of Equivalent Fluid economy	**%**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**-3.9%**	**-4.6%**	**-7.20%**	**-7.34%**	**-3.38%**	**-4.25%**	**-8.7%**	**-7.8%**
Avg.kmpl Results	**Average Fuel Economy**	**kmpl**	**6.75**		**3.73**		**6.04**		**3.53**		**6.47**		**3.45**		**5.81**		**3.24**	
	Status of Average Fuel economy	**%**	**Base**		**Base**				**Base**		**-4.05%**		**-7.49%**		**-3.78%**		**-8.26%**	
	Average Eq. Fluid Economy	**kmpl**	**6.60**		**3.61**		**5.91**		**3.41**		**6.32**		**3.35**		**5.69**		**3.13**	
	Average Eq. Fluid Economy by Lit/Ton/100 km	**Lit/Ton/100 km**	1.5153		0.7911		1.6917		0.8372		1.5820		0.8071		1.7588		0.8633	
	Status of Avg. Eq. Fluid Economy by Lit/Ton/100 km	**%**	Base		Base		Base		Base		**-4.4%**		**-2.0%**		**-4.0%**		**-3.1%**	
Oveall Results L+I	**Overall Avg. Eq. Fluid Economy by L/Ton/100 km : L + UL**	**80%** / **20%**	**0.8748**				**0.9313**				**0.8947**				**0.9612**			
	Status of Overall Avg. Eq. Fluid Economy by L/Ton/100 km(%) : L + UL	**%**	**Base**				**Base**				**-2.3%**				**-3.2%**			

Note: Equivalent Urea consumption is taken as 50% of actual urea consumption.

8.3 Tractive Effort Comparison :



8.4 Torque Table :

8.4.1: Test vehicle (Signa 3723.T BS-VI OBD-II (220 HP))

Economy mode

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2650
1	570	650	450	450	450	450	400	400	400	400	100	5
1.33	570	650	450	450	450	450	400	400	400	400	100	5
1.84	570	650	600	550	550	500	450	450	450	400	100	5
3.05	570	650	600	550	550	500	450	450	450	400	100	5
5.2	570	650	925	925	915	850	790	732	692	486	100	5
9.36	570	650	925	925	915	850	790	732	692	486	100	5

Balance mode

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2650
1	570	650	600	600	600	600	550	550	550	486	100	5
1.33	570	650	600	600	600	600	550	550	550	486	100	5
1.84	570	650	650	650	600	600	550	550	500	486	100	5
3.05	570	650	650	650	600	600	550	550	500	486	100	5
5.2	570	650	925	925	915	850	790	732	692	486	100	5
9.36	570	650	925	925	915	850	790	732	692	486	100	5

8.5 Observation: Nil

8.6 Further Testing: Further fuel economy evaluation needs to be done with benchmark vehicle to get the PALS attribute wrt core competitor.

Annexure-1

Level 1 Attribute:	PANEL Chart	VE CVBU/FE/01
PAT- Fuel Economy	Specification:Max Power- 220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-950 DD | 5.12 RAR | 295/90R 20 Tyres | iUSFE | Torque Factor: E-141% | B-131% EOL Parameters : LBSC: 2000,1900 RPM | VAM: 3.0,2.5 kmph/s @ 10,40 kmph | GDP: 50,50 kmph | RSG: 80 kmph	Panel Chart No: VE CVBU/ HCV/ SIGNA 3723.T 6.7L BSVI-II / FE/01

Programme / MY : Signa 3723.T Cx 6.7L BSVI-II	**Gateway :** DR3	**Owner:** Ajaykumar Jadhav	**Project Name: Signa 3723.T 5.6L BSVI-II**	**Date:** 16/12/2025

Level 1 Summary:

	PALS Position	Status: Current/ Gateway	Status: Job#1 Predicted
Level 1:	L	UC	

Statement Of Intent (SOI) or DNA Description:

Fuel Economy: Haulage Application

Attribute Target Summary:

Level 2 / Level 3 / Level 4 / Level 5 / Level 6	PALS	Metric	Target	Status: Current/Gateway SIGNA 3723.T BSVI-II	Status: Job#1 Predicted	Base Vehicle: Base Signa 3523.T BSVI-II	Remarks
Specification				LBSC - 2000,1900,37000 Kg VAM- 3.0 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph		LBSC - 2000,1900,35000 Kg VAM- 3.0 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph	
Vehicle GVW (Payload)		kg		37000		35000	
Power (Max) @ rpm		HP		220 hp @ 2300 rpm		220 hp @ 2300 rpm	
Power to Weight ratio		HP/Ton		5.95		6.28	
Torque (Max) @ rpm		Nm		925 Nm @ 1000-1300 rpm,		925 Nm @ 1000-1300 rpm,	
Torque to Weight ratio		Nm/Ton		25.00		26.42	
Tyres size		--		Bridgestone 295/90R20		Apollo 295/90R20	
Gear Box Ratios		--		G950 DD, FGR -9.36		G950 DD, FGR -9.36	
Final Drive Ratio		--		5.12		5.12	
Fluid Economy at Rated Load	L			UC			
Highway AC OFF		--					
Route : Jamshedpur-Ranchi		--					
Avg. Fluid Economy @40 kmph by Lit/Ton/100 km		Lit/Ton/100 km	< 0.7920	0.8070		0.7920	
Avg. Fluid Economy @55 kmph by Lit/Ton/100 km		Lit/Ton/100 km		0.8640		0.8380	
Fluid Economy At Unladen	C			UC			
Highway AC OFF		--					
Route : Jamshedpur-Ranchi		--					
Avg. Fluid Economy @40 kmph by Lit/Ton/100 km		Lit/Ton/100 km	< 1.5150	1.5810		1.5150	*Signa 3723.T BSVI-II and Cowl 3523.T BSVI-II was tested back to back condition 0n JSR-Ranchi route. *Signa 3723.T BSVI-II is with Non-THU bearing, with Non low CRR Tyre. FLE is calculated by Lit/ton/100 km method because of incresed GVW, Where lower value is better.
Avg. Fluid Economy @55 kmph by Lit/Ton/100 km		Lit/Ton/100 km		1.7590		1.6920	
Overall Fuel Economy (80% Laden + 20% Unladen)	L			UC			
Highway AC OFF		--					
Route : Jamshedpur-Ranchi		--					
Overall Avg. Fluid Economy @40 kmph by Lit/Ton/100 km		Lit/Ton/100 km	< 0.8760	0.8950		0.8760	
Overall Avg. Fluid Economy @55 kmph by Lit/Ton/100 km		Lit/Ton/100 km		0.9620		0.9320	

Executive Summary:

1. Test vehicle Signa 3723.T BSVI-II tested with : 220 HP Engine calibration with Non- THU Bearing + Low viscous 15W50 engine oil + Low viscous 80W90 rear axle oil + Low idle 650 RPM + Bridgestone Non- LCRR tyres + 24" Fan + Modified USFE table.
2. Target has been taken at-par wrt to base Signa 3523.T BSVI-II vehicle.
3. Signa 3723.T BSVI-II is not meeting PALS target of "L".
4. FLE is calculated by Lit/ton/100 kms method because of incresed GVW, Where lowest value is better.

Tested by:	Verified and Approved by:
Palash Das / Swapnil Jadhav **PAT ENGINEER**	Ajaykumar Jadhav / Sanjeev Kumar **PAT LEAD / PROJECT LEAD**

Annexure-2

TATA MOTORS LTD. ERC,PUNE	PAT ATTRIBUTES SHEET - Fuel Economy		VE CVBU/FE/01
			PAT Sheet No.: VE CVBU/ HCV/ SIGNA 3723.T 5.6L BSVI RDE/ FE/01
	Depart ment Name: Vehicle Engineering	**Project Name: SIGNA 3723.T 5.6L BSVI-II (G-950 DD FGR-9.36, 5.12 RAR, 295/90R 20 phase-I Tyres, iUSFE mode)** **Base Vehicle: Cowl 3523.T 5.6L BSVI-II_G-950 DD, FGR-9.36, 5.12 RAR, 295/90R 20 Tyres_iUSFE** **Application: Haulage**	**Date:** 16/12/2025

Sr.No.	Aggregate and test parameter						Unit	Test Type	Classification @DR4	Ref.Standard / Work Guideline	Simulation	Base Vehicle	Target	Test Vehicle	Status (RYG)	Remarks	CoC Remark
	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7						SIGNA 3523.T BSVI OBD-II (G 950 DD | RAR 5.12 | 295/90R20)		SIGNA 3723.T BSVI OBD-II (G 950 DD | RAR 5.12 | 295/90R20)			
A	Fuel Economy Evaluation								Heading			LBSC - 2000,1900,35000 Kg VAM- 3.0 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph		LBSC - 2000,1900,37000 Kg VAM- 3.0 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph	--		
2.0		Fuel Economy At Rated Load							Heading								
2.7			Highway AC OFF						Heading								
2.7.1				Route : Jamshedpur-Ranchi				Standard	Specification	TST/VT/WG/158						*Signa 3723.T BSVI-II and Signa 3523.T BSVI-II was tested back to back condition 0n JSR-Ranchi route. *Signa 3723.T BSVI-II is with Non-THU bearing, with Non low CRR Tyre, FLE is calculated by Lit/ton/100 km method because of incresed GVW, Where lowest value is better.	
2.7.12				Average Eq. Fluid Economy @40 kmph by Lit/Ton/100 km			Lit/Ton/100 km		Target			0.7920	< 0.7920	0.8070	R		
2.7.13				Average Eq. Fluid Economy @55 kmph by Lit/Ton/100 km			Lit/Ton/100 km		Data Genration			0.8380		0.8640	--		
4.0		Fuel Economy At Unladen							Heading								
4.7			Highway AC OFF						Heading								
4.7.1				Route : Jamshedpur-Ranchi					Specification	TST/VT/WG/158						--	
4.7.11				Average Eq. Fluid Economy @40 kmph by Lit/Ton/100 km			Lit/Ton/100 km	Standard	Target			1.5150	< 1.5150	1.5810	R		
4.7.12				Average Eq. Fluid Economy @55 kmph by Lit/Ton/100 km			Lit/Ton/100 km		Data Genration			1.6920		1.7590	--		
5.0		Overall Fuel Economy (80% Laden + 20% Unladen)							Heading								
5.1			Highway AC OFF						Heading								
5.1.1				Route : Jamshedpur-Ranchi					Specification	TST/VT/WG/158						--	
				Overall (L + UL) Avg. Eq. Fluid Economy @40 kmph by L/Ton/100 km			Lit/Ton/100 km		Target			0.8760	< 0.8760	0.8950	R		
				Overall (L + UL) Avg. Eq. Fluid Economy @55 kmph by L/Ton/100 km			Lit/Ton/100 km		Data Genration			0.9320		0.9620	--		

Sign Off			Index for Status
Responsibility	**Name**	**Signature**	Off target, no resolution
PAT Engineer	**Palash Das/ Swapnil Jadhav**		Marginally off target, resolution planned/low risk
PAT Lead / Project Lead	**Ajaykumar Jadhav / Sanjeev kumar**		Met target, no concern

Fuel Economy Evaluation on Signa 3723.T BSVI-II with Cowl 3523.T BSVI-II Base Vehicle.

PED - HCV I 16.12.2025

OUR CULTURE CREDO

AT TATA MOTORS
We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.
We are **MORE WHEN ONE!**

Vehicle Specification

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Specification		Test Vehicle	Base Vehicle
Vehicle	Model	Signa 3723.T Atulya BSVI-II	Signa 3523.T Ash BSVI-I
	GVW (Rated load) Kg	37000(Kg)	35000(kg)
	GVW (Unladen) kg	10,080	10,020
Engine	Emission norm	BSVI RDE	BSVI RDE
	Engine	Ashwamedh BSVI-II 5.6L	Ashwamedh BSVI-II 5.6L
	Calibration	95%	As per production released
	Technology	--	--
	Torque	925 Nm @ 1000-1300 rpm	925 Nm @ 1000 - 1300 rpm
	Power	220 HP @ 2300 rpm	220 HP @ 2300 rpm
	Torque to weight ratio	25	26.42
	Power to weight ratio	5.95	6.28
Driveline	Gear box	G950 6S DD, FGR- 9.36	G950 6S DD, FGR- 9.36
	Gear 1	9.36	9.36
	Gear 2	5.2	5.2
	Gear 3	3.05	3.05
	Gear 4	1.84	1.84
	Gear 5	1.33	1.33
	Gear 6	1	1
	RAR	5.12	5.12
	Tyre	295/90R20	295/90R20

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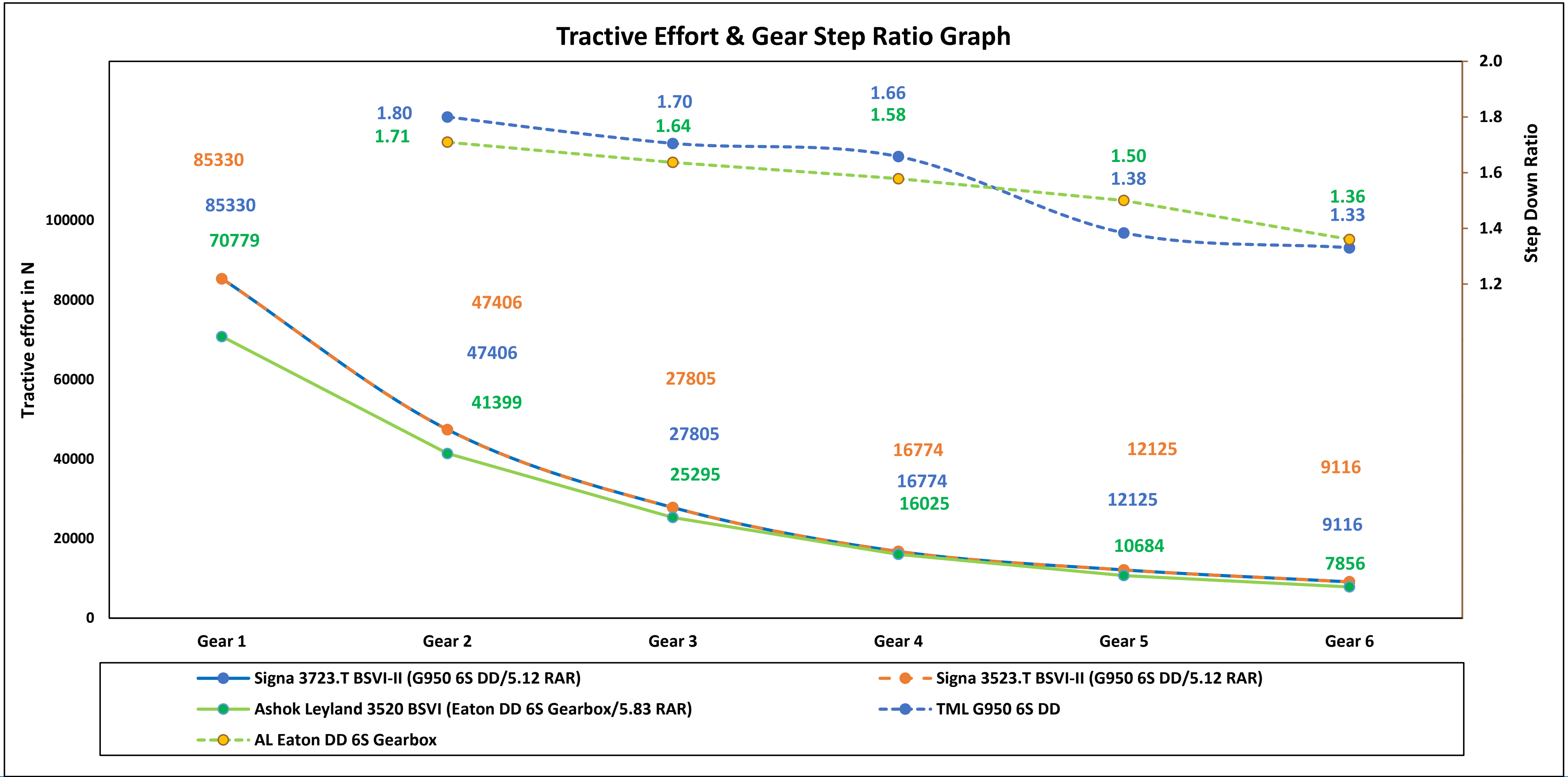
Tractive Effort Comparison

TATA MOTORS
Connecting Aspirations



- Tractive effort of Signa 3723.T Cx BSVI-II vehicle is better as compared with Signa 3523.T Ash. BSVI-I in all gear.

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FLE Summary Status (Laden & Unladen)

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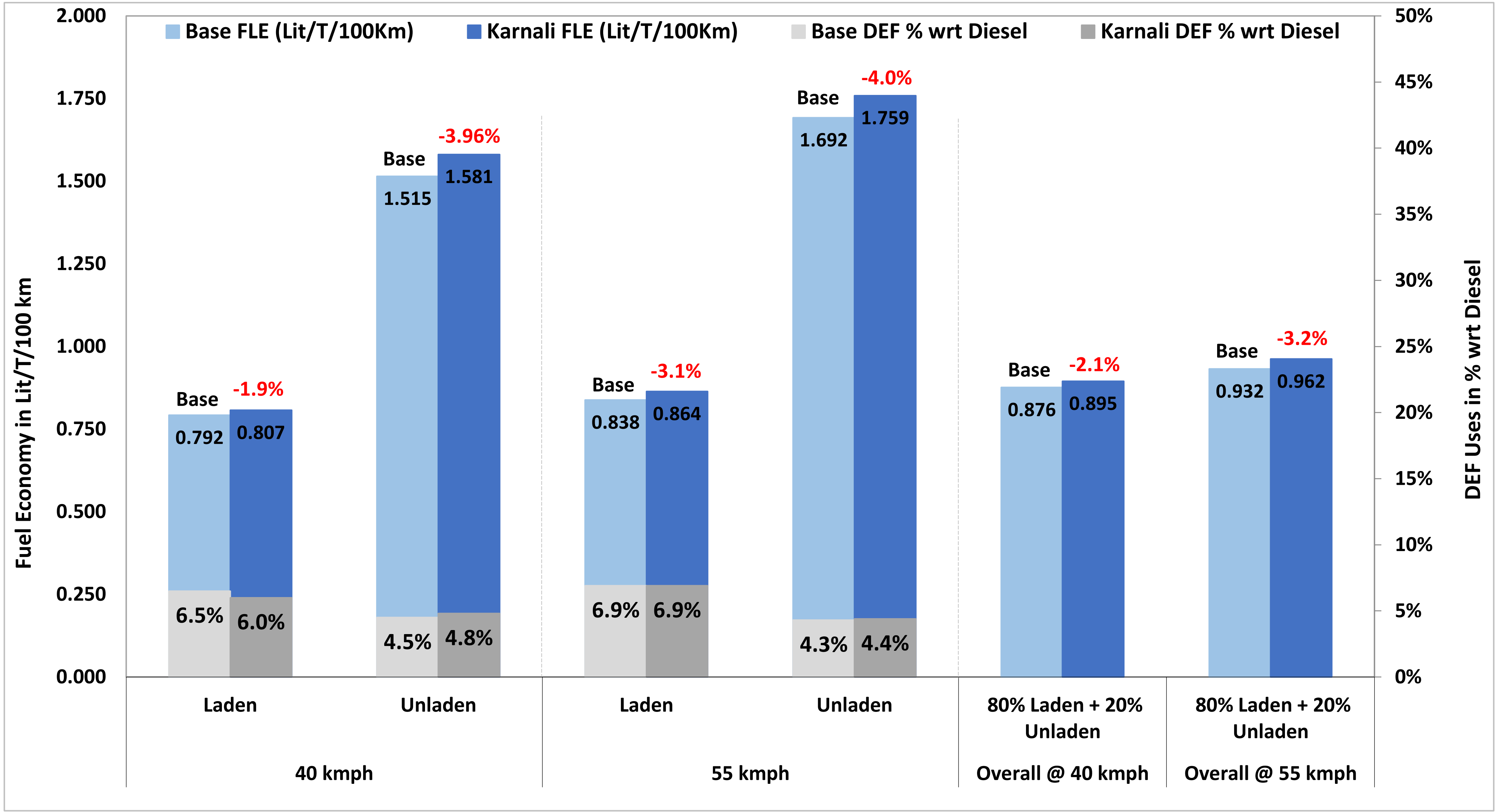

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Duty Cycle Analysis

Jamshedpur-Ranchi Road FE Trial

Duty Cycle : Summary (Laden)

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JSR-RANCHI-JSR Duty Cycle					
		07.12.2025 (40 kmph)		10.12.2025 (55 kmph)	
Parameters		Signa 3723.T RDE Test Vehicle	Signa 3523.T BSVI Base Vehicle	Signa 3723.T RDE Test Vehicle	Signa 3523.T BSVI Base Vehicle
Engine idle speed (rpm)		**650**	**650**	**650**	**650**
Low idle time (%)		5.64	5.80	6.00	8.18
Average engine speed (rpm)		1068.30	1064.62	1228.85	1205.89
Average power (kW)		40.04	36.68	47.74	43.23
Avg. vehicle speed (kmph)	Including Stop	36.19	36.25	40.89	40.66
	Excluding Stop	38.49	38.46	43.73	44.33
FE switch utilization (%time)	Power	17.12	16.90	16.53	8.43
	Balance	0.0	0.0	0.0	0.0
	Eco	82.88	83.10	83.47	67.78
Gear utilization (%time)	Neutral Gear	5.4	3.8	6.6	6.6
	4th Gear	3.3	4.5	2.8	3.2
	5th Gear	11.35	10.60	9.68	15.67
	6th Gear	69.85	70.58	66.42	84.33
	Total shifts	287.00	244.00	244.00	195.00
Throttle (%time)	100%	14.5	15.3	12.6	14
	0%	35.2	39.8	32.4	38.5
Coolant Temp (°C)	80°C	**0.2**	14.1	**0.2**	8
	85°C	**1.7**	46.9	**1.5**	52.7
	90°C	**52.2**	23.8	**59.1**	27
SCRTM Fuel spent (%)		--	--	--	--
Regeneration Fuel spent (%)		--	--	--	--
Duty Cycle fuel economy (kmpl)		**3.68**	**4.04**	**3.50**	**3.84**
Duty cycle Distance(km)		247.54	249.46	238.21	237.56
Duty Cycle Time(Hrs)		**6.84**	**6.88**	**5.83**	**5.84**
Actual Avg. Fuel Economy		**3.78**	**3.57**	**3.61**	**3.42**

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Duty Cycle : Summary (Unladen)

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JSR-RANCHI-JSR Duty Cycle		14.12.2025 (40 kmph)		12.12.2025 (55 kmph)	
Parameters		Signa 3723.T RDE Test Vehicle	Signa 3523.T BSVI Base Vehicle	Signa 3723.T RDE Test Vehicle	Signa 3523.T BSVI Base Vehicle
Engine idle speed (rpm)		**650**	**650**	**650**	**650**
Low idle time (%)		3.3	3.04	6.38	5.39
Average engine speed (rpm)		1000.55	1013.67	1176.83	1189.47
Average power (kW)		17.24	19.03	24.42	18.42
Avg. vehicle speed (kmph)	Including Stop	36.18	36.78	41.84	42.38
	Excluding Stop	38.45	38.93	46.35	47.43
FE switch utilization (%time)	Power	0.00	0.00	0.00	0.00
	Balance	0.00	0.00	0.00	0.00
	Eco	100	100	100	100
Gear utilization (%time)	Neutral Gear	4	3.3	7.5	6
	4th Gear	1.5	1.4	1.1	1.5
	5th Gear	6.00	6.00	6.19	5.74
	6th Gear	76.94	77.04	86.73	82.79
	Total shifts	140.00	125.00	140.00	125.00
Throttle (%time)	100%	0.1	1.1	0.4	2.9
	0%	11.9	14.9	13.8	18.4
Coolant Temp (°C)	80°C	**0.3**	14.4	**0.9**	2.4
	85°C	**1.9**	82.4	**1.3**	90.1
	90°C	**93.7**	1.7	**93.2**	4.3
SCRTM Fuel spent (%)		--	--	--	--
Regeneration Fuel spent (%)		0.00	0.00	0.00	0.00
Duty cycle fuel economy (kmpl)		**6.03**	**6.35**	**6.9**	**6.61**
Duty cycle Distance(km)		**244.45**	**246.57**	**243.28**	**244.34**
Duty Cycle Time(Hrs)		**6.53**	**5.82**	**5.09**	**5.77**
Actual Avg. Fuel Economy		**6.6**	**5.79**	**6.07**	**6.37**

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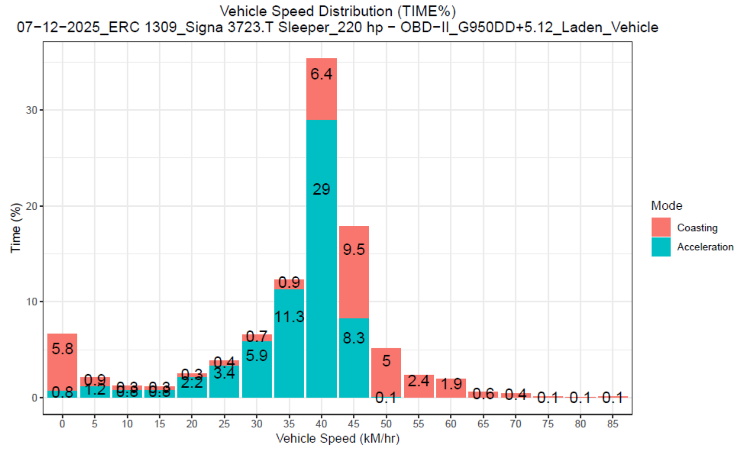
Vehicle Speed Distribution

Laden Trial

TATA MOTORS
Connecting Aspirations

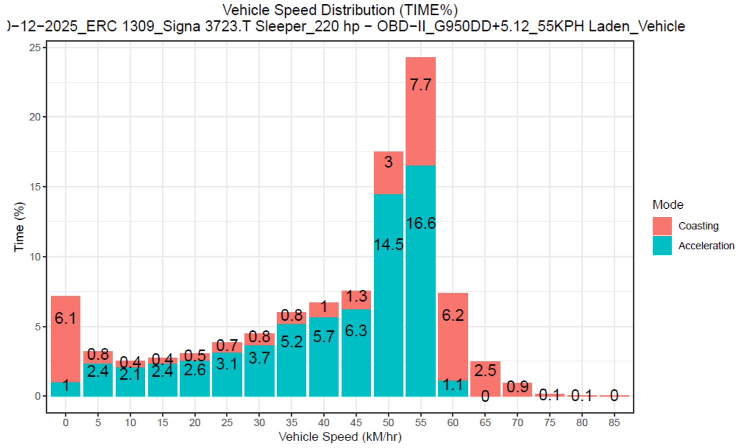
Signa 3723.T RDE Test Vehicle

07.12.25 **40 kmph**



Avg Vehicle Speed :- 36.19 Avg Vehicle Speed(excluding Stop) :- 38.49 Max Vehicle Speed :- 85.98 Coasting :- 36.06 % Acceleration :- 63.94 %

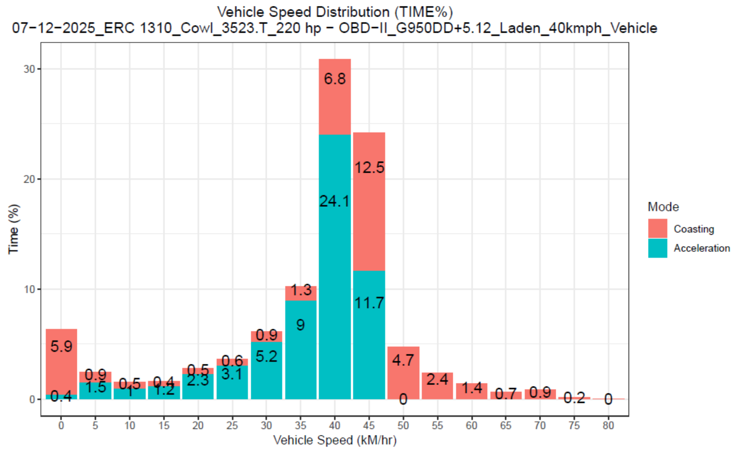
10.12.25 **55 kmph**



Avg Vehicle Speed :- 40.89 Avg Vehicle Speed(excluding Stop) :- 43.73 Max Vehicle Speed :- 84.25 Coasting :- 33.41 % Acceleration :- 66.59 %

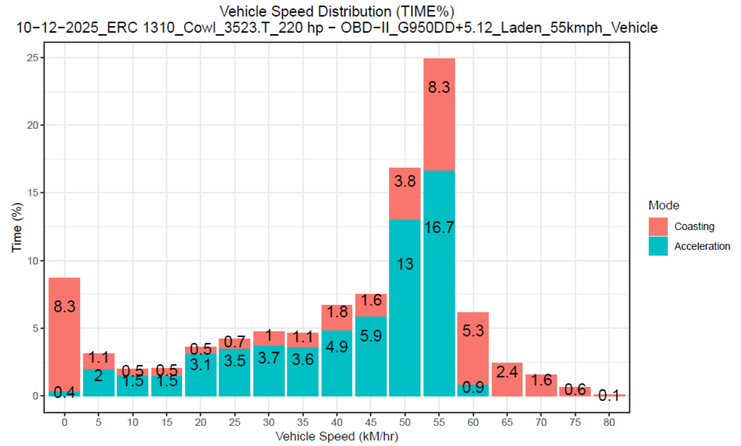
Signa 3523.T Base Vehicle

07.12.25 **40 kmph**



Avg Vehicle Speed :- 36.25 Avg Vehicle Speed(excluding Stop) :- 38.46 Max Vehicle Speed :- 78.38 Coasting :- 40.51 % Acceleration :- 59.49 %

10.12.25 **55 kmph**



Avg Vehicle Speed :- 40.66 Avg Vehicle Speed(excluding Stop) :- 44.33 Max Vehicle Speed :- 80.94 Coasting :- 39.31 % Acceleration :- 60.69 %

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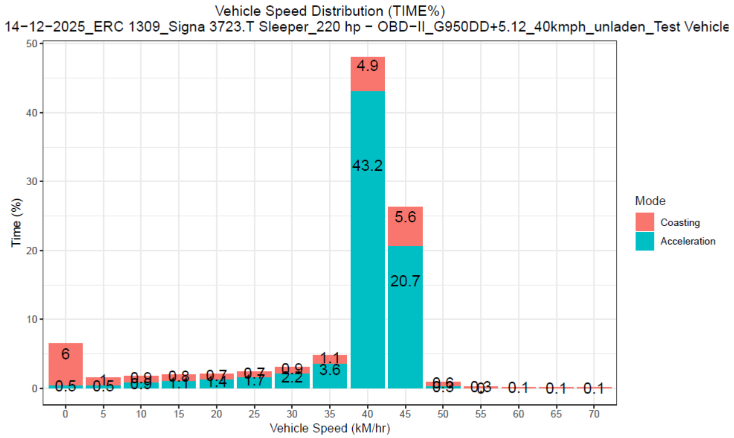
Vehicle Speed Distribution

Unladen Trial

TATA MOTORS
Connecting Aspirations

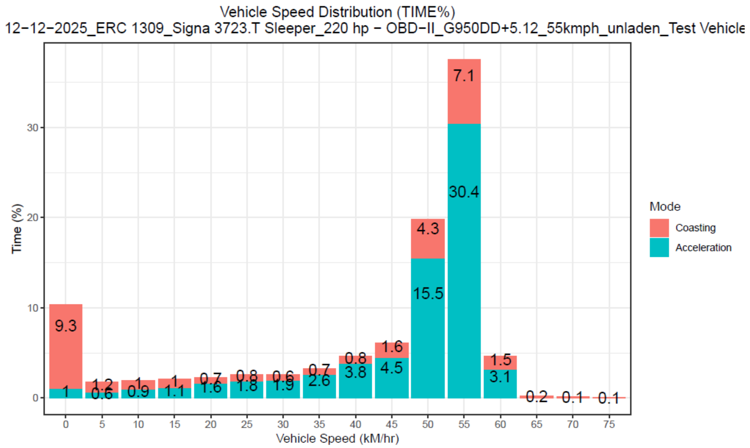
Signa 3723.T RDE Test Vehicle

14.12.25 40 kmph



Avg Vehicle Speed :- 36.18 Avg Vehicle Speed(excluding Stop) :- 38.45 Max Vehicle Speed :- 71.63 Coasting :- 23.89 % Acceleration :- 76.11 %

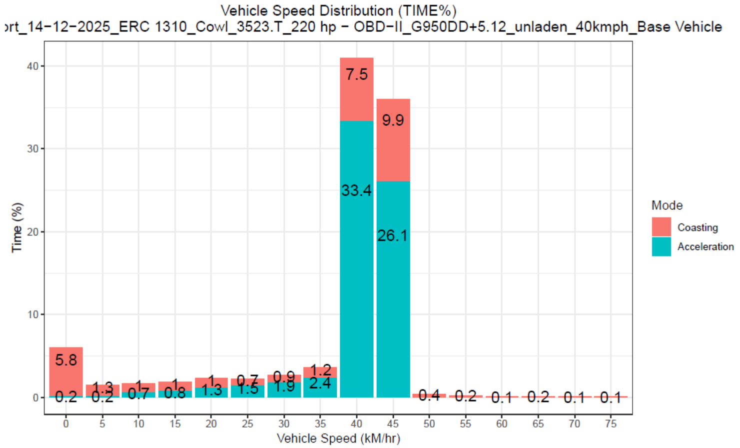
12.12.25 55 kmph



Avg Vehicle Speed :- 41.84 Avg Vehicle Speed(excluding Stop) :- 46.35 Max Vehicle Speed :- 74.22 Coasting :- 31.01 % Acceleration :- 68.99 %

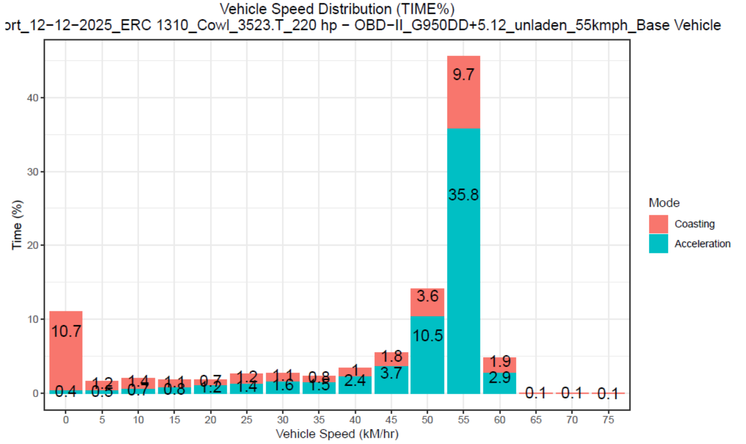
Signa 3523.T Base Vehicle

14.12.25 40 kmph



Avg Vehicle Speed :- 36.78 Avg Vehicle Speed(excluding Stop) :- 38.93 Max Vehicle Speed :- 76.38 Coasting :- 31.44 % Acceleration :- 68.56 %

12.12.25 55 kmph



Avg Vehicle Speed :- 42.38 Avg Vehicle Speed(excluding Stop) :- 47.43 Max Vehicle Speed :- 75.34 Coasting :- 36.62 % Acceleration :- 63.38 %

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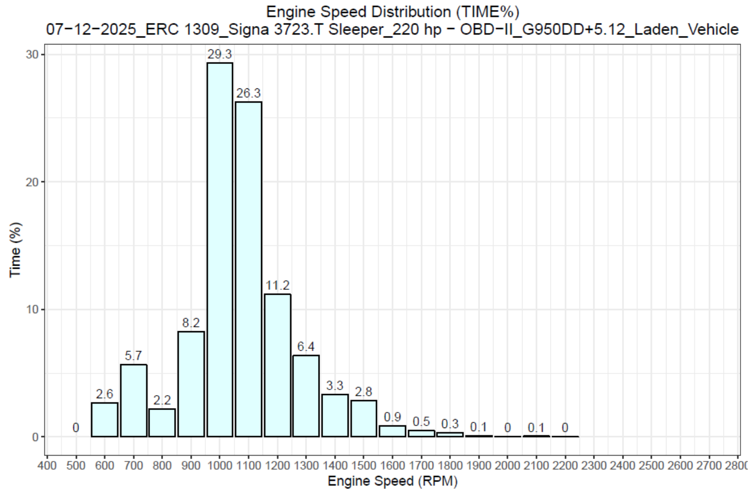
Engine Speed Distribution

Laden Trial

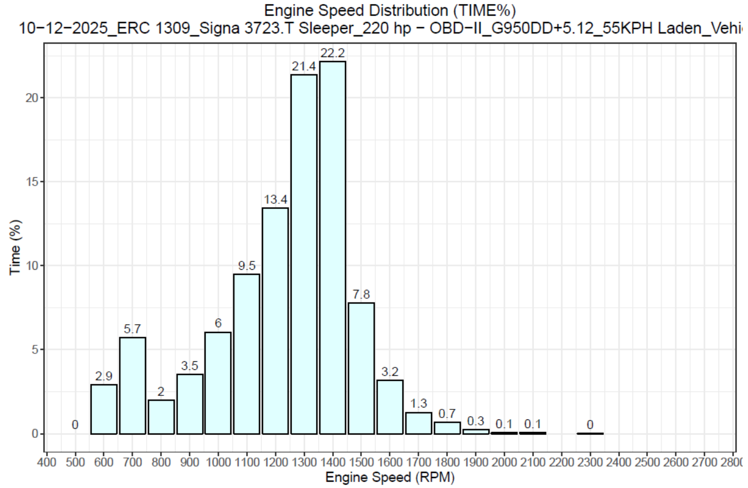
TATA MOTORS
Connecting Aspirations

Signa 3723.T RDE Test Vehicle

07.12.25 **40 kmph**

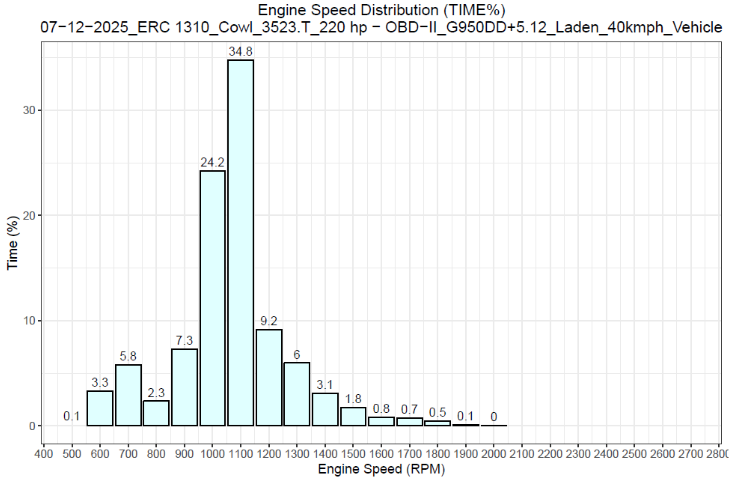


10.12.25 **55 kmph**

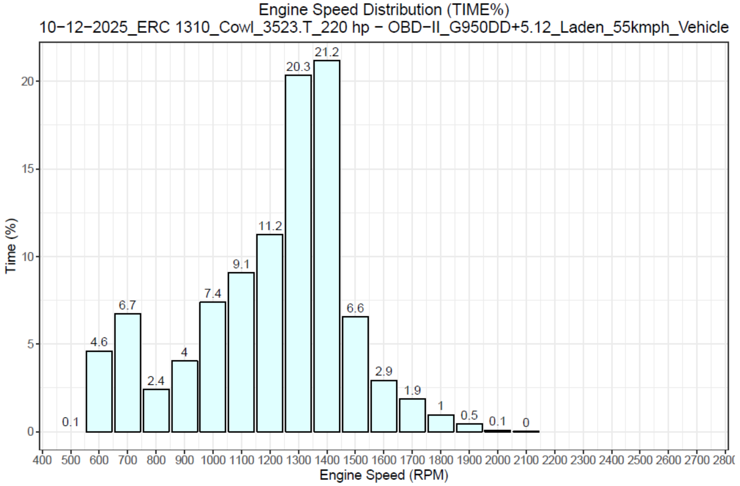


Signa 3523.T Base Vehicle

07.12.25 **40 kmph**



10.12.25 **55 kmph**



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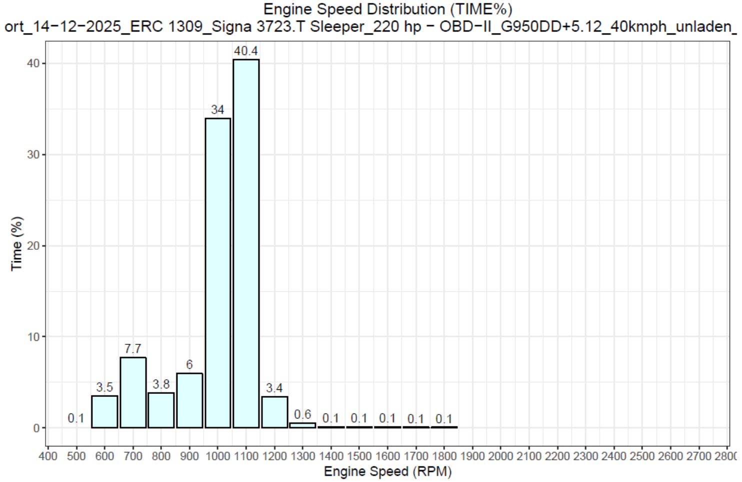
Engine Speed Distribution

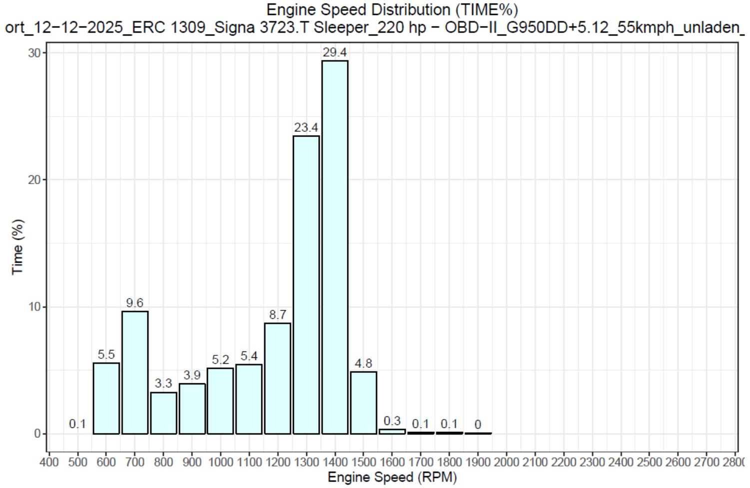
Unladen Trial

Signa 3723.T RDE Test Vehicle

14.12.25 **40 kmph**

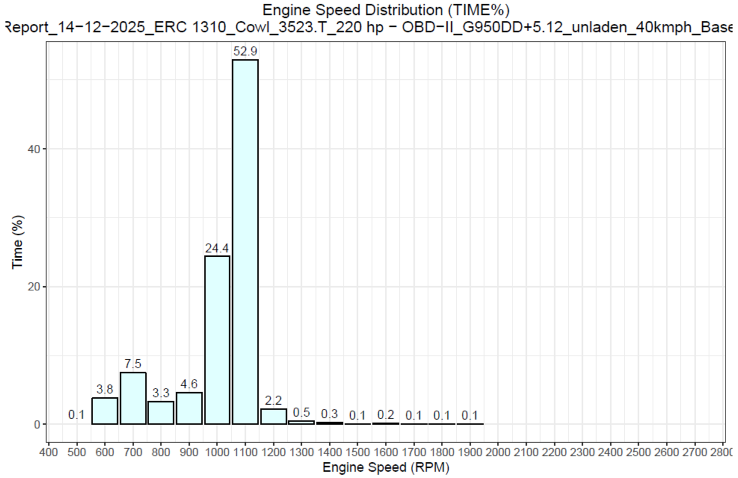


12.12.25 **55 kmph**

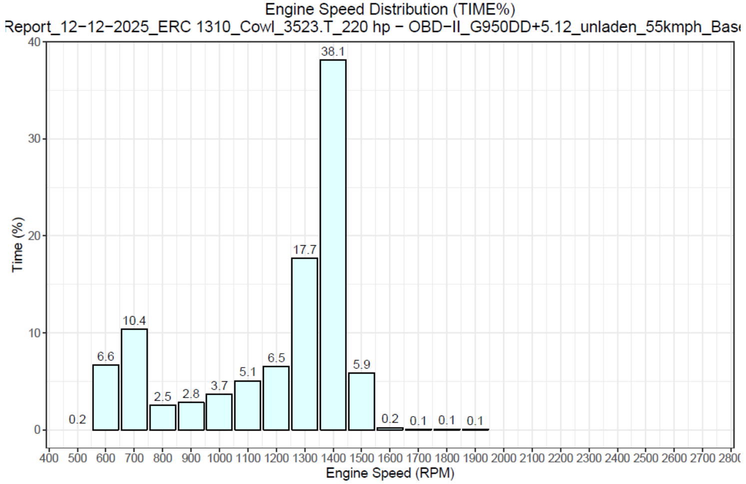


Signa 3523.T Base Vehicle

14.12.25 **40 kmph**



12.12.25 **55 kmph**



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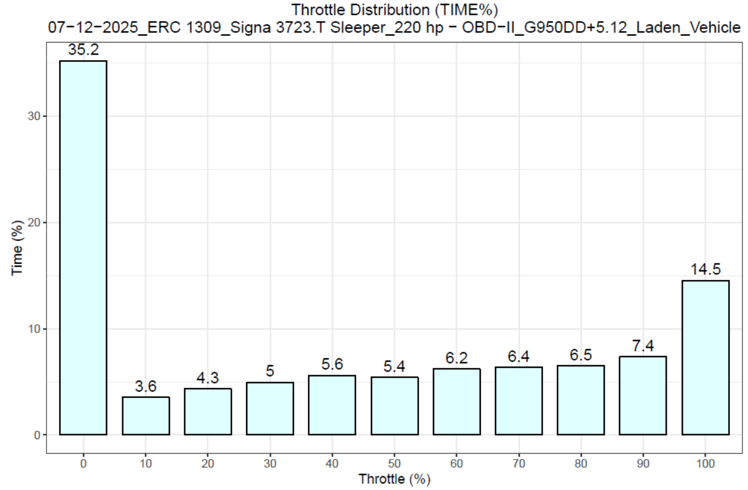
Throttle Distribution

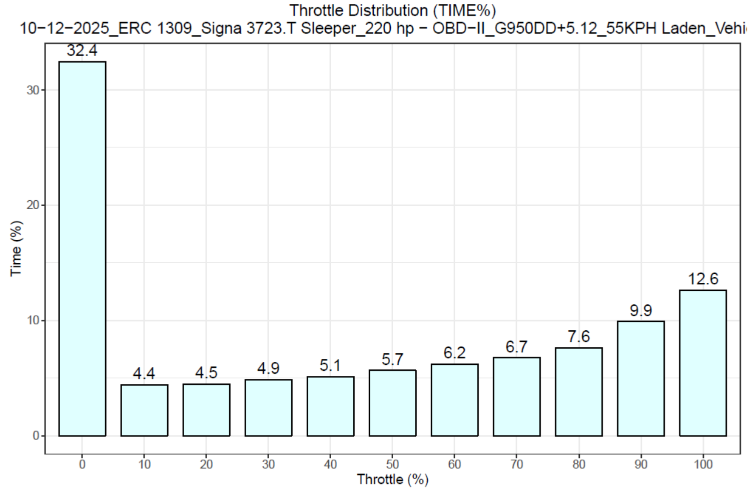
Laden Trial

Signa 3723.T RDE Test Vehicle

07.12.25 **40 kmph**

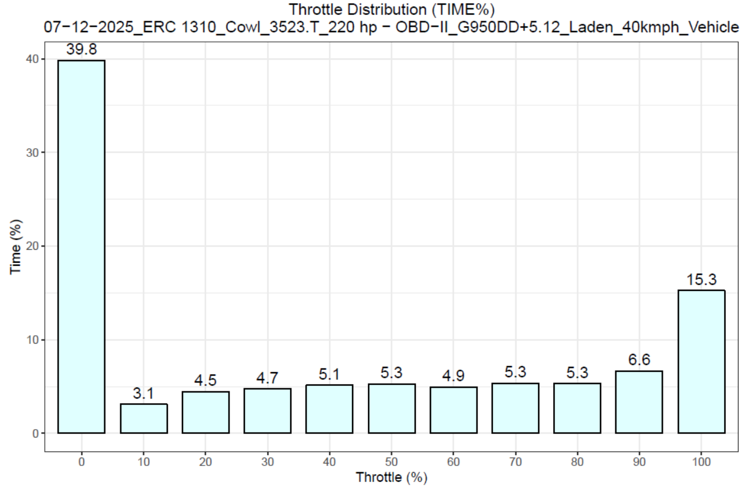


10.12.25 **55 kmph**

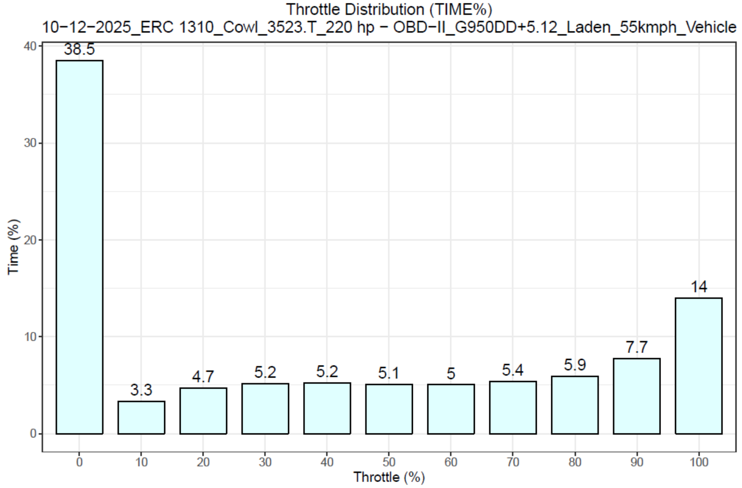


Signa 3523.T Base Vehicle

07.12.25 **40 kmph**



10.12.25 **55 kmph**



Throttle Distribution

Unladen Trial

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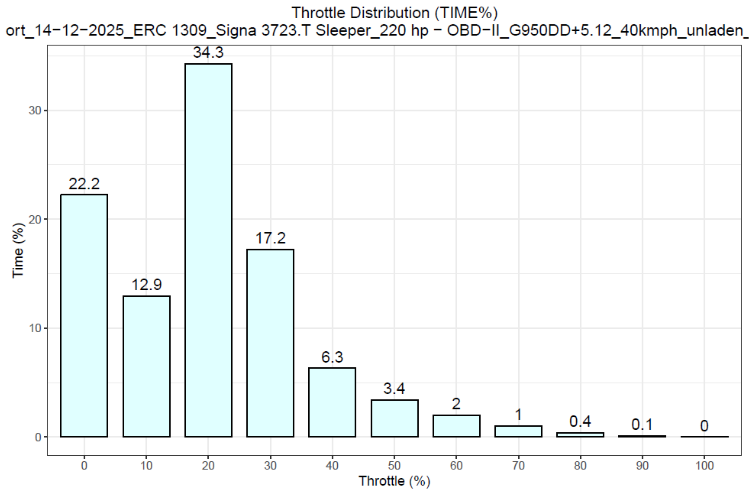
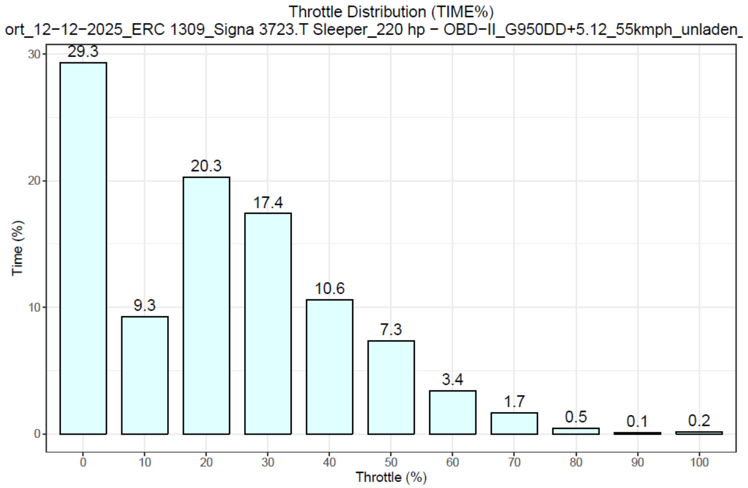
Signa 3723.T RDE Test Vehicle

14.12.25 | **40 kmph**

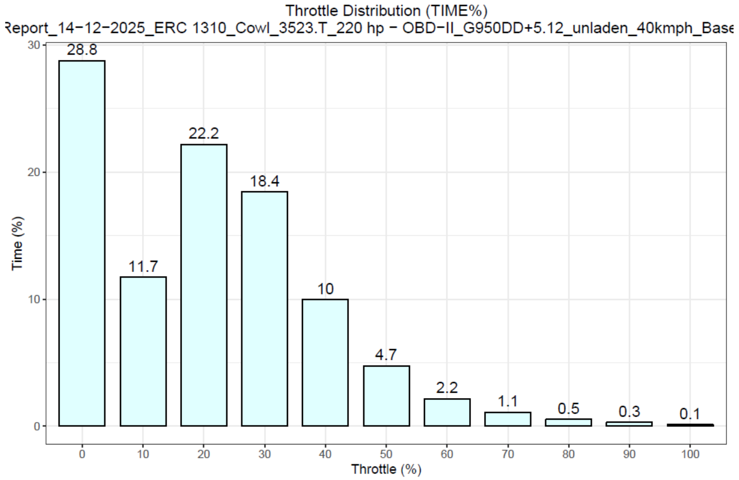


12.12.25 | **55 kmph**

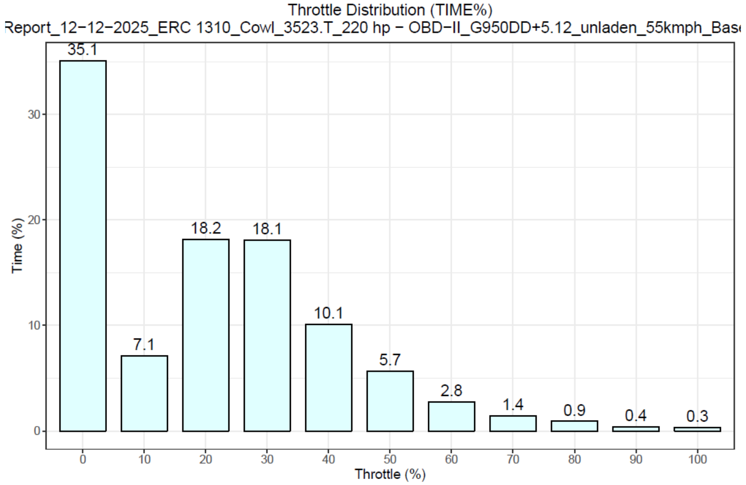


Signa 3523.T Base Vehicle

14.12.25 | **40 kmph**



12.12.25 | **55 kmph**



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Gear Utilization

Laden Trial

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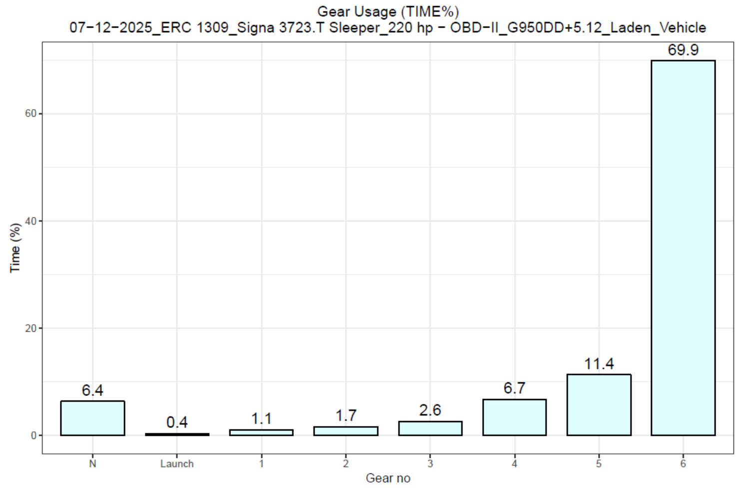
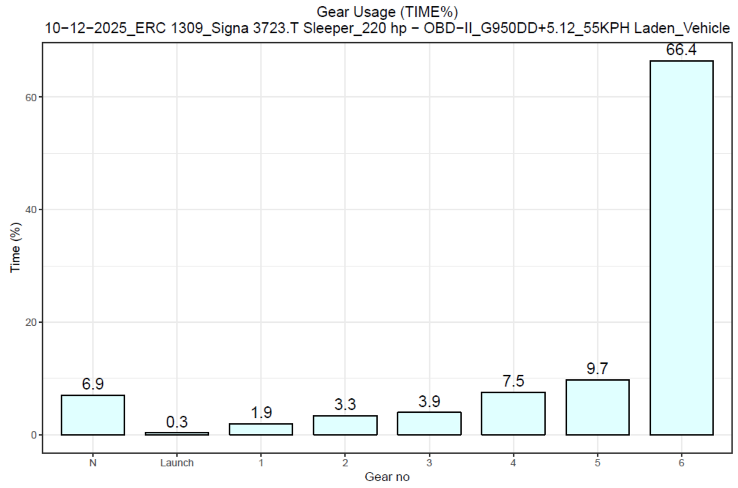
Signa 3723.T RDE Test Vehicle

07.12.25 **40 kmph**

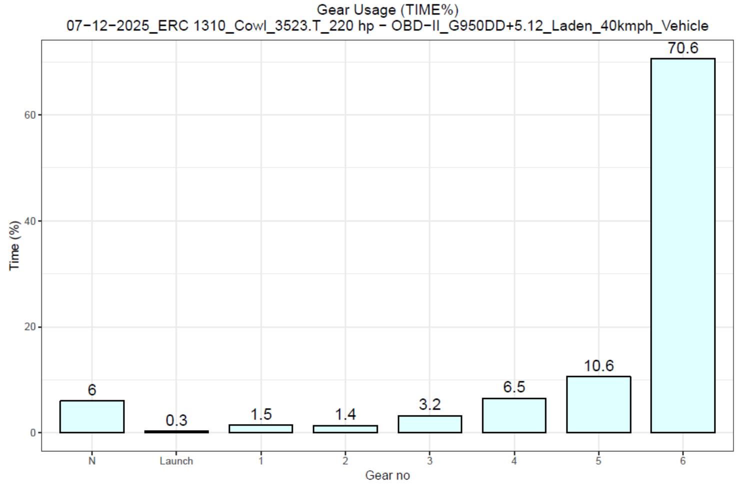


10.12.25 **55 kmph**

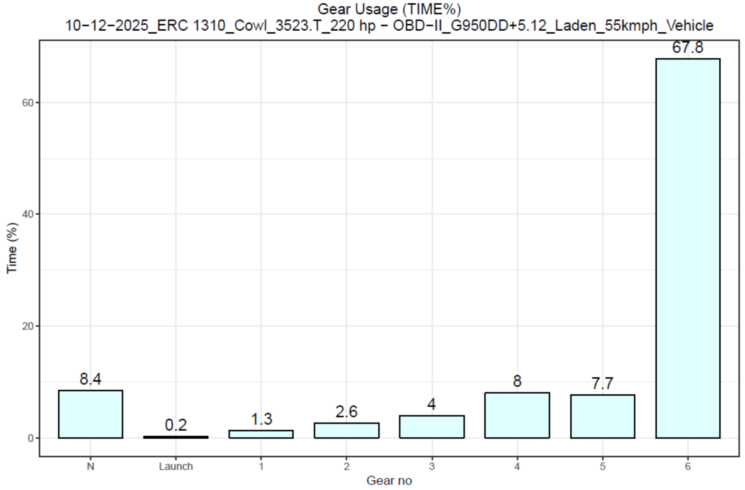


Signa 3523.T Base Vehicle

07.12.25 **40 kmph**



10.12.25 **55 kmph**



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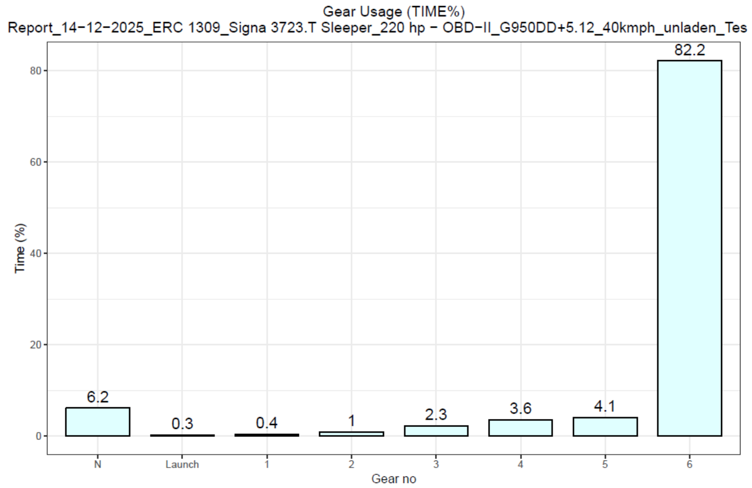
Gear Utilization

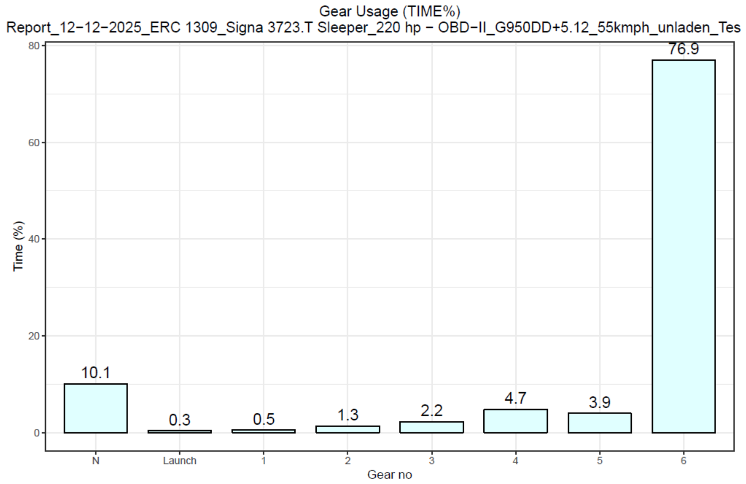
Unladen Trial

Signa 3723.T RDE Test Vehicle

14.12.25 **40 kmph**

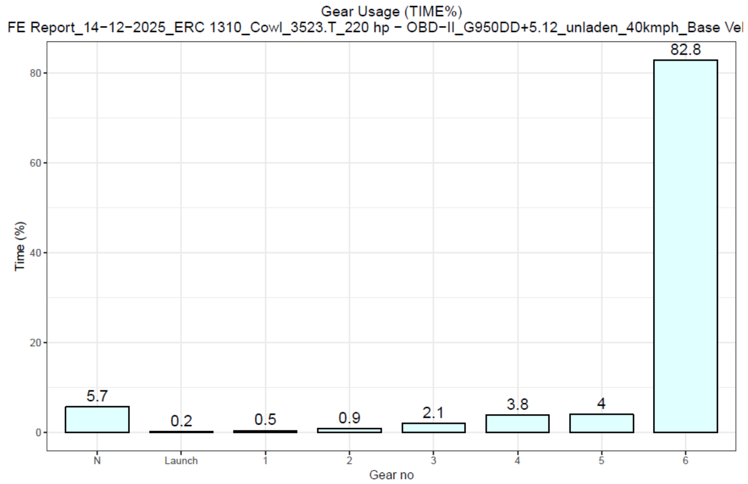


12.12.25 **55 kmph**

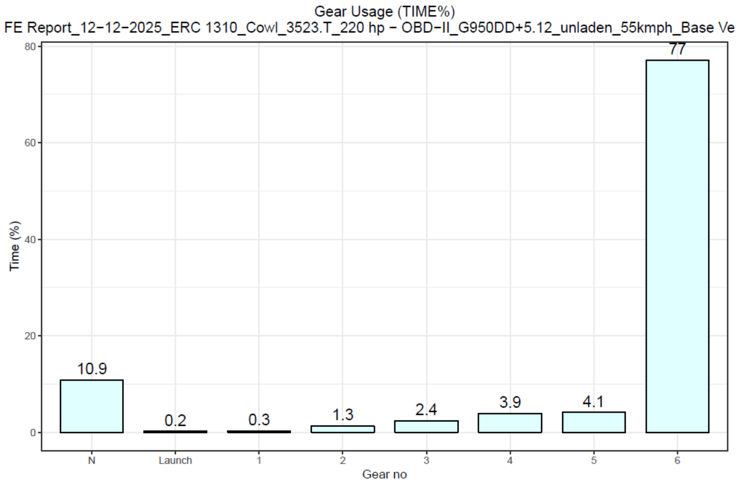


Signa 3523.T Base Vehicle

14.12.25 **40 kmph**



12.12.25 **55 kmph**



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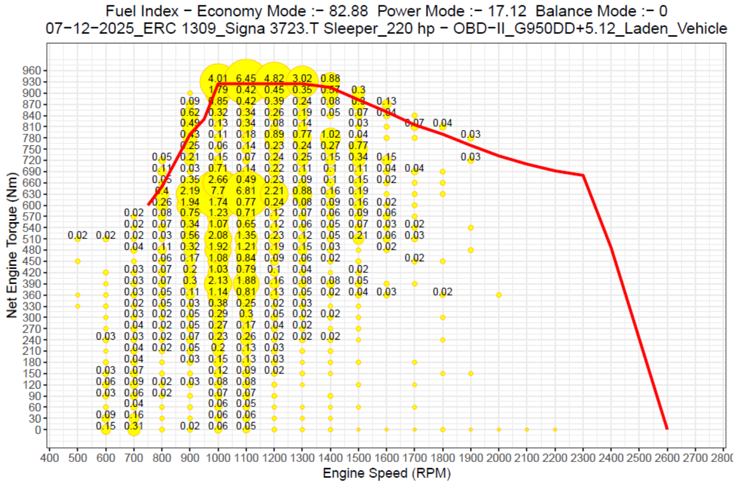
Fuel spent % as Engine speed & torque

TATA MOTORS
Connecting Aspirations

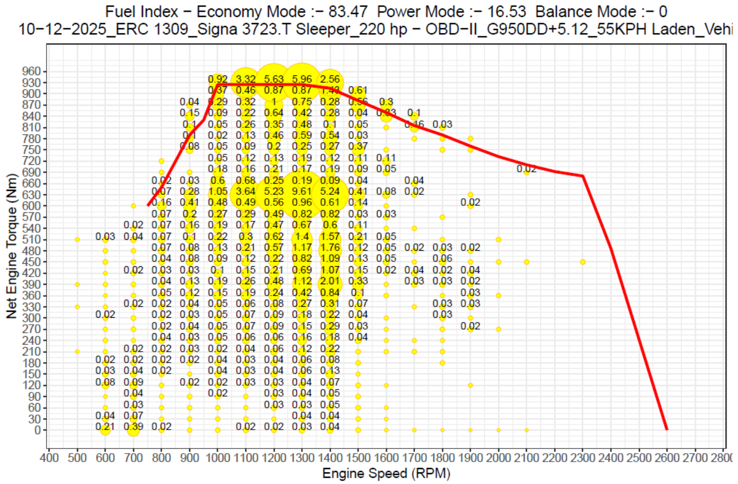
Laden Trial

Signa 3723.T RDE Test Vehicle

07.12.25 **40 kmph**

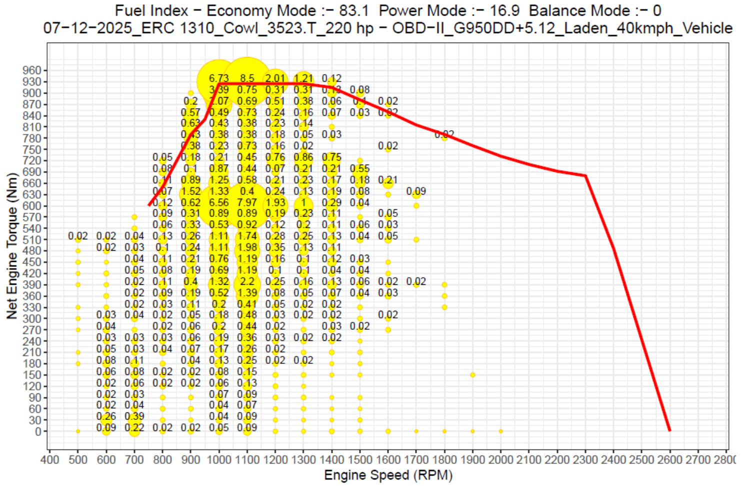


10.12.25 **55 kmph**

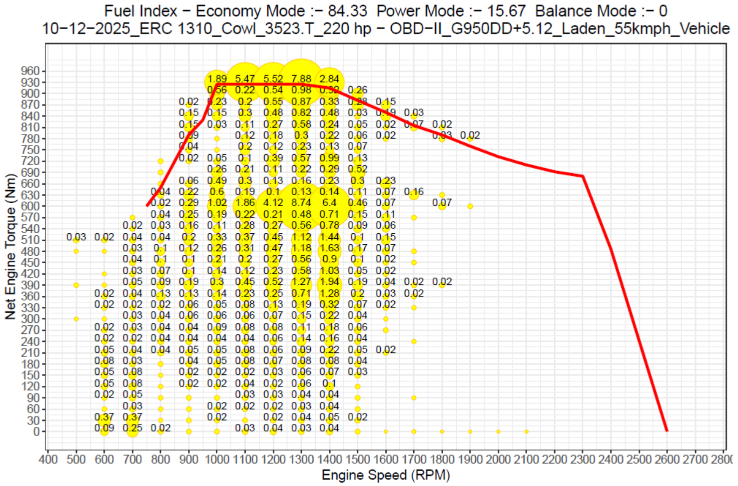


Signa 3523.T Base Vehicle

07.12.25 **40 kmph**



10.12.25 **55 kmph**



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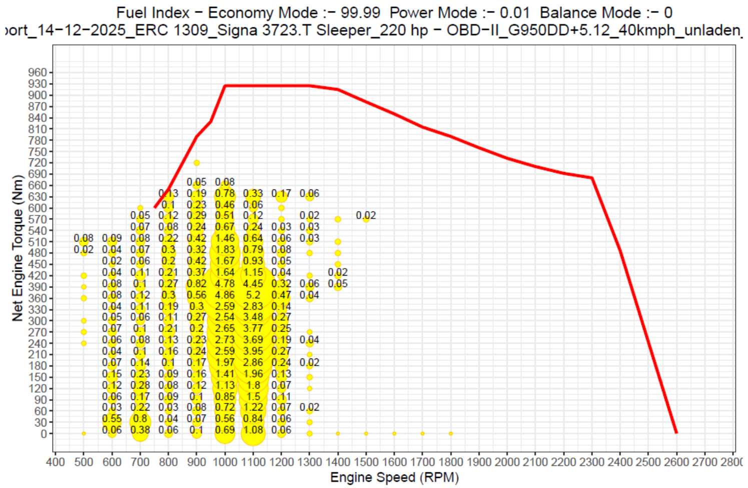
Fuel spent % as Engine speed & torque

TATA MOTORS
Connecting Aspirations

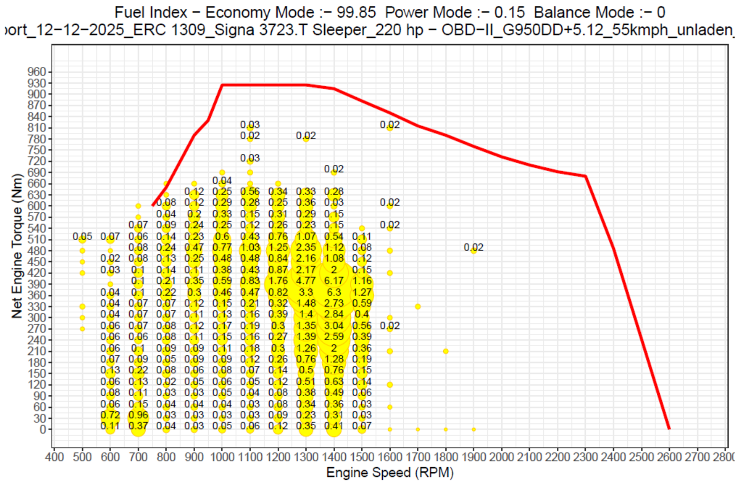
Unladen Trial

Signa 3723.T RDE Test Vehicle

14.12.25 **40 kmph**

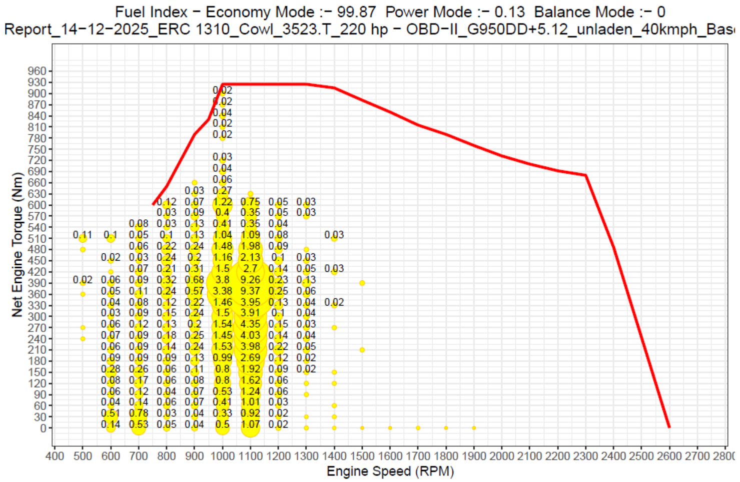


12.12.25 **55 kmph**

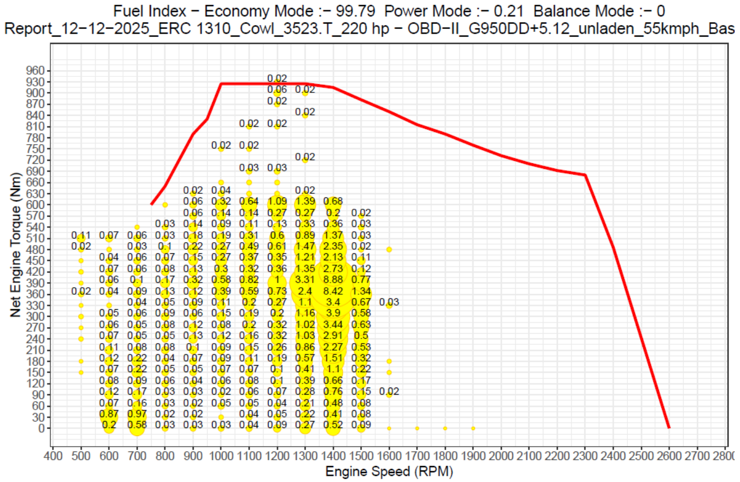


Signa 3523.T Base Vehicle

14.12.25 **40 kmph**



12.12.25 **55 kmph**



MOREWHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

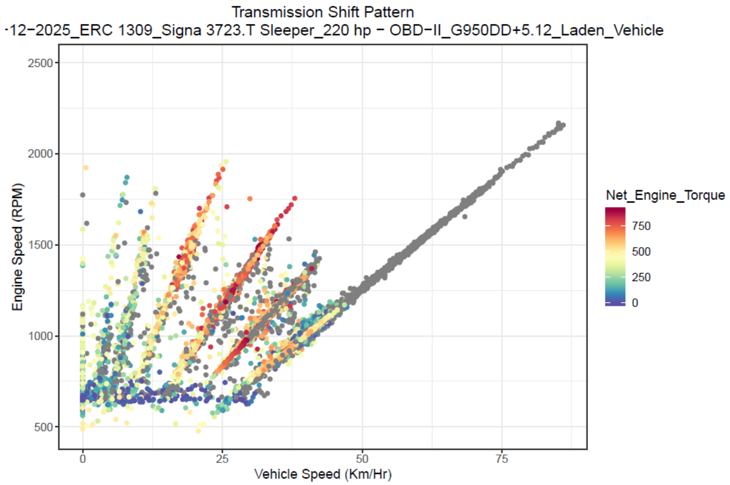
Transmission Shift Pattern

Laden Trial

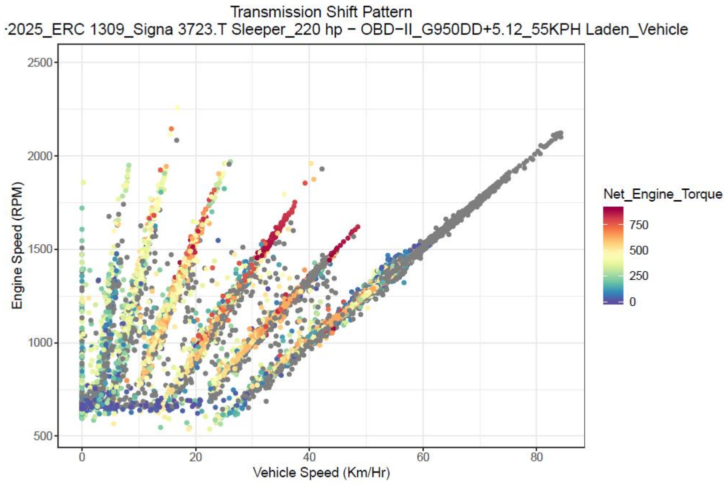
TATA MOTORS
Connecting Aspirations

Signa 3723.T RDE Test Vehicle

07.12.25 **40 kmph**

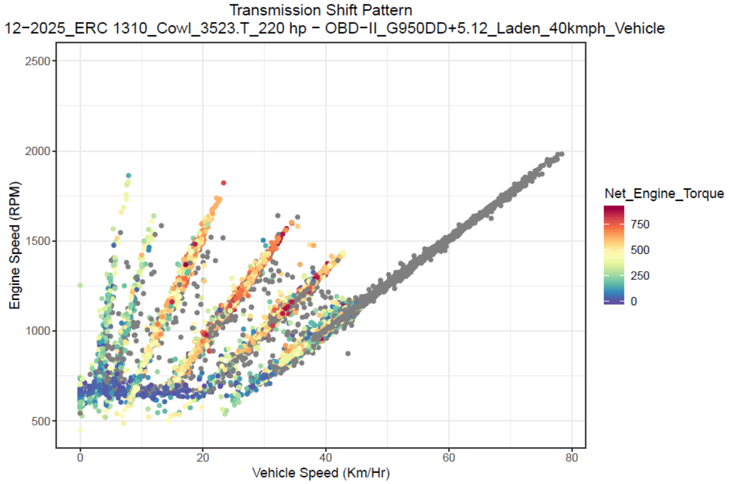


10.12.25 **55 kmph**

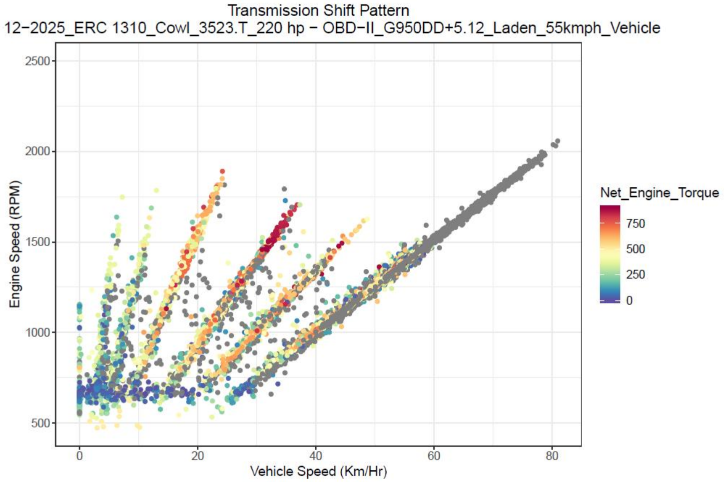


Signa 3523.T Base Vehicle

07.12.25 **40 kmph**



10.12.25 **55 kmph**



MOREWHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

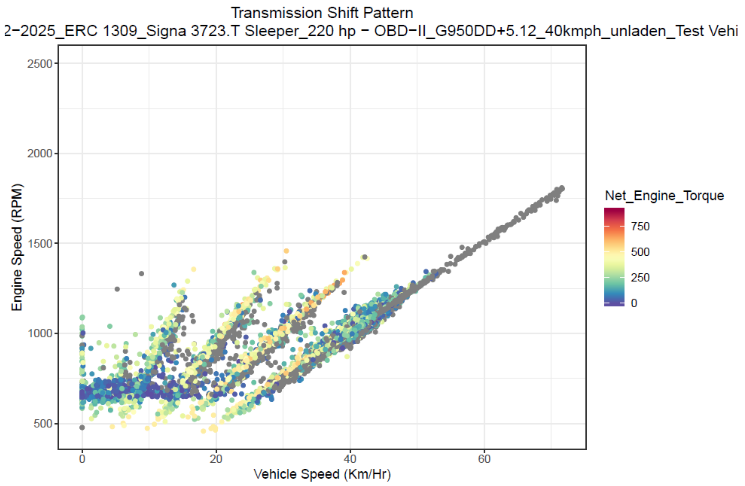
Transmission Shift Pattern

Unladen Trial

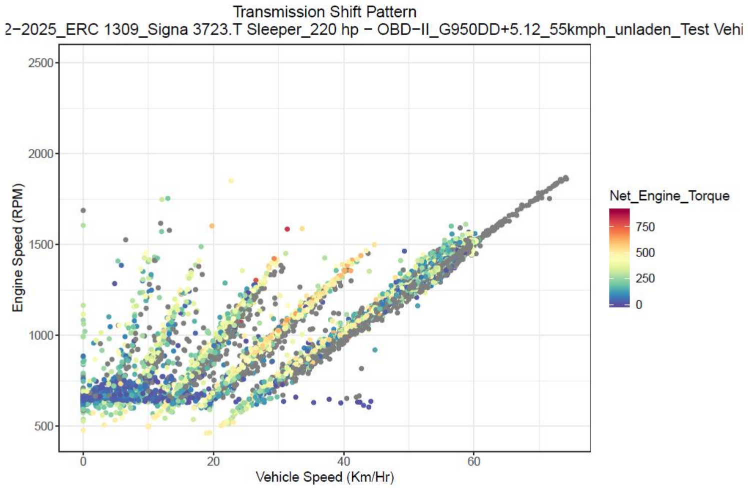
TATA MOTORS
Connecting Aspirations

Signa 3723.T RDE Test Vehicle

14.12.25 **40 kmph**

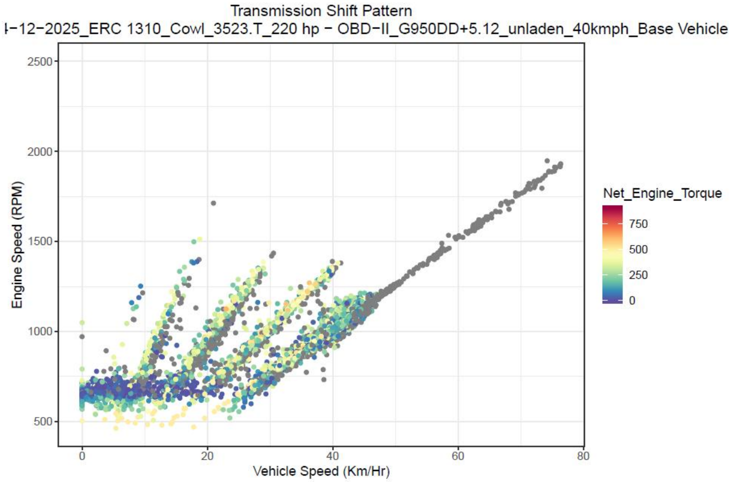


12.12.25 **55 kmph**

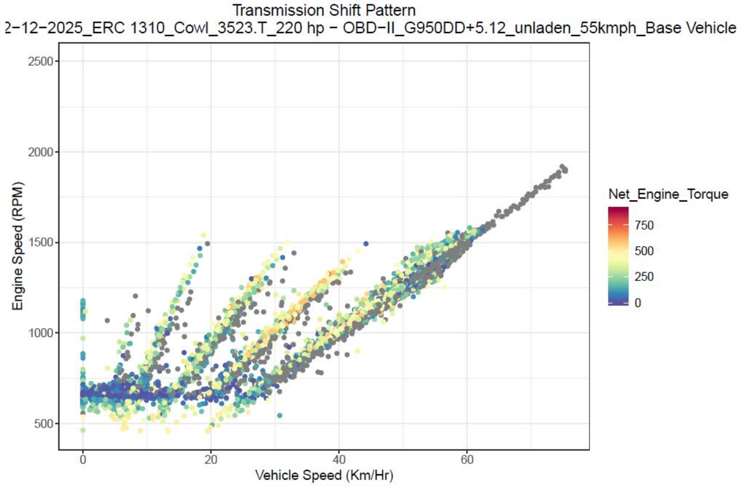


Signa 3523.T Base Vehicle

14.12.25 **40 kmph**



12.12.25 **55 kmph**



MOREWHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

Inference

Fluid Economy(FLE) Status of 3723.T BSVI RDE Test vehicle w.r.t. 3523.T BSVI **base vehicle :**

1. Laden

a) 40kmph :- Fluid Economy of 3723.T by Lit/Ton /100 km is **inferior** than base vehicle by 1.9 % .

b) 55kmph :- Fluid Economy of 3723.T by Lit/Ton /100 km is **inferior** than base vehicle by 3.1 % .

2. Unladen

a) 40kmph :- Fluid Economy of 3723.T by Lit/Ton /100 km is **inferior** than base vehicle by 4.36 %.

b) 55kmph :- Fluid Economy of 3723.T by Lit/Ton /100 km is **inferior** than base vehicle by 4.0 %.

Overall Average Fluid Economy(FLE) Status of 3723.T BSVI RDE Test vehicle w.r.t. 3523.T BSVI base vehicle :

a) 40kmph (Laden(80%) +Unladen(20%)):- Fluid Economy of 3723.T by Lit/Ton /100 km is **inferior** than base vehicle by **2.2 %** .

b) 55kmph (Laden(80%) +Unladen(20%)):- Fluid Economy of 3723.T by Lit/Ton /100 km is **inferior** than base vehicle by **3.2 %** .

Thank You

TATA MOTORS LIMITED ERC PUNE	TEST PLAN	DATE: 19/01/2026			
	VEHICLE TESTING DEPARTMENT				
Description: Report on FE evaluation of Signa 3723.T BSVI-II 5.6L, G950 (FGR:9.36) and 5.12 RAR in comparison with Cowl 3723.T BSVI-II veh.		**Test Item No:** UA002748			
		New	✓	**Moderate**	x
		Major	x	**Minor**	x

Part No: 52070261R,	**Modification No:** NR;3,	**Enclosures:** FIR attached
Drawing No: 52070261R		
Make: -	**WBS Number:** P.21.EA.456- Shuriken Karnali (Axle Spacing Program)	**Copies to** :
Vehicle Type: Request from Mr.Bimal Kishor dated 05/01/2023	**No. of Samples:** -	
Test Requested by:		**Equipments:**
Test Requisition Reference:	Testing-Outdoor-UA/02766	
Reason for Test/s:	Data Generation	
Regulatory Requirement:	a) Component: As per work guideline	
	b) Vehicle: As per work guideline	

REMARKS: As per FIR	**INTERNAL REF:**	NA
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		

PREPARED BY	CHECKED BY	APPROVED BY
Palash Das		

TEST REQUISITION

Ref.	-	Date:	18/12/2025

To,
ERC Testing
Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 3723.T BSVI				
2	Part Number	52070261R,				
3	Drawing number	52070261R	Mod status	NR;3,	Mod Date	15/12/2025
4	Vehicle Model / Project	Request from Mr.Bimal Kishor dated 05/01/2023				
5	WBS Number	P.21.EA.456Shuriken Karnali (Axle Spacing Program)				
6	Supplier	-				
7	Number of samples	-				
8	Inspection report	Enclosed	✔	If No commitment date:		
9	Material report	Enclosed	✔	If No commitment date:		
10	Suppliers test report	Enclosed	✔	If No commitment date:		
11	Reason for testing	First Source Development	✘	Weight Reduction	✘	
		Alternate source development	✘	Cost Reduction	✘	
		Data generation Test	✔	Regulatory test	✘	
		Any other (pl. specify)	✘	New Development	✘	
12	What testing required?					
13	Remarks					
14	Peripheral Parts Availability	No				

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

Authorized Signature:	
Name:	
Dept:	

Contact Person:	Subhankar Banerjee
Phone Number	